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THESIS

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Theme

Digital Twin for eHealth: State of the Art and Conceptual Architecture

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Abstract

The digital transformation of healthcare has led to the emergence of connected medical devices, electronic health records, remote monitoring platforms, and artificial intelligence-based decision support systems. Although these technologies generate large amounts of heterogeneous data, many eHealth systems still remain fragmented, static, and weakly personalized. Digital Twin technology offers a promising paradigm to address these limitations by creating dynamic virtual representations of physical entities, processes, or patients, continuously updated through data exchange and supported by simulation, analytics, and decision services.

This Master thesis presents a structured state-of-the-art literature review of Digital Twins for eHealth. It first introduces Digital Twin fundamentals, including the distinction between digital models, digital shadows, Digital Twins, and intelligent Digital Twins, before presenting the eHealth and digital-health environment in which these principles can be applied. It then analyzes healthcare-oriented Digital Twin applications, architectures, enabling technologies, and implementation challenges. Cardiovascular Digital Twins are used as a representative clinical deep-dive because they combine biosignals, imaging, hemodynamic measurements, longitudinal monitoring, and patient-specific modeling. Particular attention is given to interoperability, data governance, privacy, security, explainability, validation, and ethical considerations. The literature synthesis derives a general multi-layer eHealth Digital Twin architecture, together with data-flow, cross-cutting requirement, and evaluation findings. The final chapter concludes the Master study and introduces CardioTwin as a practical cardiovascular continuation for the PFE, while the thesis remains centered on Digital Twins for eHealth in a general healthcare context.

Keywords: Digital Twin, eHealth, Digital Health, Healthcare, IoT, IoMT, Artificial Intelligence, Simulation, Interoperability, Clinical Decision Support, Cardiovascular Digital Twin.

Résumé

La transformation numérique du secteur de la santé a favorisé l'apparition des dispositifs médicaux connectés, des dossiers médicaux électroniques, des plateformes de télésurveillance et des systèmes d'aide à la décision basés sur l'intelligence artificielle. Malgré la quantité importante de données générées, plusieurs systèmes eHealth restent fragmentés, statiques et faiblement personnalisés. La technologie du jumeau numérique constitue une approche prometteuse pour répondre à ces limites, en créant des représentations virtuelles dynamiques d'entités physiques, de processus ou de patients, mises à jour par des échanges de données et enrichies par la simulation, l'analyse et les services décisionnels.

Ce mémoire de Master présente une revue structurée de l'état de l'art sur les jumeaux numériques appliqués à l'eHealth. Il introduit d'abord les fondements du jumeau numérique, notamment la distinction entre modèle numérique, ombre numérique, jumeau numérique et jumeau numérique intelligent, avant de présenter l'environnement eHealth et santé numérique dans lequel ces principes peuvent être appliqués. Ensuite, il analyse les applications, architectures, technologies habilitantes et défis d'implémentation des jumeaux numériques dans le domaine de la santé. Les jumeaux numériques cardiovasculaires sont utilisés comme un approfondissement clinique représentatif, car ils combinent biosignaux, imagerie, mesures hémodynamiques, suivi longitudinal et modélisation spécifique au patient. Une attention particulière est accordée à l'interopérabilité, à la gouvernance des données, à la confidentialité, à la sécurité, à l'explicabilité, à la validation et aux considérations éthiques. La synthèse de la littérature permet de dégager une architecture générale multicouche pour les jumeaux numériques eHealth, ainsi que des principes liés au flux de données, aux exigences transversales et à l'évaluation. Le dernier chapitre conclut l'étude de Master et introduit CardioTwin comme une continuation cardiovasculaire pratique pour le PFE, tout en conservant l'orientation générale du mémoire sur les jumeaux numériques pour l'eHealth.

Mots-clés : Jumeau numérique, eHealth, santé numérique, IoT, IoMT, intelligence artificielle, simulation, interopérabilité, aide à la décision clinique, jumeau numérique cardiovasculaire.

الملخص

أدت التحولات الرقمية في قطاع الصحة إلى ظهور الأجهزة الطبية المتصلة، والسجلات الصحية الإلكترونية، ومنصات المراقبة عن بعد، وأنظمة دعم القرار المعتمدة على الذكاء الاصطناعي. ورغم أن هذه التقنيات تنتج كميات كبيرة من البيانات المتنوعة، فإن العديد من أنظمة الصحة الإلكترونية ما تزال مجزأة، وثابتة، ومحدودة التخصيص. وتقدم تقنية التوأم الرقمي مقاربة واعدة لمعالجة هذه الحدود، من خلال إنشاء تمثيلات افتراضية ديناميكية للكيانات الفيزيائية أو العمليات أو المرضى، يتم تحديثها باستمرار عبر تبادل البيانات، وتدعيمها بالمحاكاة والتحليل وخدمات دعم القرار.

يعرض هذا البحث مراجعة منظمة لحالة الفن حول توظيف التوائم الرقمية في مجال الصحة الإلكترونية. ويتناول أولا أسس التوأم الرقمي، بما في ذلك التمييز بين النموذج الرقمي، والظل الرقمي، والتوأم الرقمي، والتوأم الرقمي الذكي، ثم يقدم بيئة الصحة الإلكترونية والصحة الرقمية التي يمكن تطبيق هذه المبادئ داخلها. بعد ذلك، يحلل تطبيقات التوائم الرقمية الموجهة للرعاية الصحية، ومعمارياتها، والتقنيات الممكنة لها، وتحديات تنفيذها. وتستعمل التوائم الرقمية القلبية الوعائية كحالة سريرية ممثلة لأنها تجمع بين الإشارات الحيوية، والتصوير الطبي، والقياسات الهيموديناميكية، والمتابعة الطولية، والنمذجة الخاصة بالمريض. كما يولي البحث اهتماما خاصا لقضايا قابلية التشغيل البيني، وحوكمة البيانات، والخصوصية، والأمن، وقابلية التفسير، والتحقق، والاعتبارات الأخلاقية. وتستخلص مراجعة الأدبيات معمارية عامة متعددة الطبقات للتوائم الرقمية في الصحة الإلكترونية، إضافة إلى مبادئ تتعلق بتدفق البيانات، والمتطلبات العابرة للطبقات، والتقييم. أما الفصل الأخير فيلخص دراسة الماستر ويقدم مشروع CardioTwin كاستمرار عملي في المجال القلبي الوعائي ضمن مشروع نهاية الدراسة، مع الحفاظ على الطابع العام للبحث حول التوائم الرقمية للصحة الإلكترونية.

الكلمات المفتاحية: التوأم الرقمي، الصحة الإلكترونية، الصحة الرقمية، الرعاية الصحية، إنترنت الأشياء، إنترنت الأشياء الطبية، الذكاء الاصطناعي، المحاكاة، قابلية التشغيل البيني، دعم القرار السريري، التوأم الرقمي القلبي الوعائي.

ACRONYMS

AFDB	Atrial Fibrillation Database
AI	Artificial Intelligence
API	Application Programming Interface
CDR	Clinical Data Repository
CT	Computed Tomography
CVD	Cardiovascular Disease
DL	Deep Learning
DT	Digital Twin
DT4H	Digital Twin for Health
eHealth	Electronic Health
ECG	Electrocardiogram
EHR	Electronic Health Record
EMR	Electronic Medical Record
FHIR	Fast Healthcare Interoperability Resources
HF	Heart Failure
HL7	Health Level Seven
ICT	Information and Communication Technology
INCART	St. Petersburg Institute of Cardiological Technics 12-lead Arrhythmia Database
IoMT	Internet of Medical Things
IoT	Internet of Things
MIT-BIH	Massachusetts Institute of Technology–Beth Israel Hospital Arrhythmia Database
ML	Machine Learning
MRI	Magnetic Resonance Imaging
PFE	Projet de Fin d'Etudes
PTB-XL	Physikalisch-Technische Bundesanstalt Extra Large ECG Dataset
SVDB	Supraventricular Arrhythmia Database

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Part I

General Introduction

CHAPTER 1

GENERAL INTRODUCTION

1.1 Context

Healthcare systems are increasingly supported by digital technologies such as connected medical devices, electronic health records, telemedicine platforms, mobile health applications, and artificial intelligence-based analytics. This evolution has contributed to the emergence of eHealth, a broad domain that aims to improve healthcare delivery, monitoring, prevention, diagnosis, and patient engagement through information and communication technologies.

At the same time, healthcare data are becoming more heterogeneous and continuous. A patient may generate information through clinical consultations, laboratory tests, medical imaging, wearable devices, mobile applications, and hospital information systems. The challenge is no longer only to collect data, but also to transform these data into meaningful, reliable, and actionable knowledge for healthcare professionals and patients.

Digital Twin technology has recently gained attention as a promising paradigm for representing physical entities and processes through dynamic virtual counterparts. In healthcare, a Digital Twin may support the representation of a patient, a medical device, a hospital service, a clinical workflow, or even a population-level process. The objective is to connect real-world data with virtual models in order to monitor current states, simulate possible futures, and support decision-making.

1.2 Problem Statement

Despite the progress of eHealth systems, several limitations remain. First, medical data are often distributed across multiple systems and formats, which makes integration and interoperability difficult. Second, many digital health applications provide static analysis based on isolated measurements rather than dynamic representations that evolve with the real-world situation. Third, decision support systems frequently rely on general population models and may not sufficiently account for individual variability, contextual information, and temporal evolution.

These limitations raise the following central problem:

> How can Digital Twin technology be used as a conceptual and architectural foundation for building eHealth systems that are more dynamic, interoperable, personalized, and capable of supporting monitoring, simulation, and decision-making?

1.3 Research Motivation

The motivation of this thesis comes from the need to move from fragmented eHealth applications toward integrated and intelligent healthcare ecosystems. A Digital Twin-based approach can provide a structured way to combine real-world data, virtual representation, simulation, artificial intelligence, and decision services. This is particularly relevant in healthcare because clinical decisions depend not only on current measurements, but also on patient history, context, uncertainty, and possible future trajectories.

While this thesis examines Digital Twins for eHealth in a general healthcare context, cardiovascular Digital Twins are used as a representative clinical deep-dive. This domain combines multimodal biosignals, imaging, hemodynamic measurements, longitudinal monitoring, and patient-specific physiological modeling, making it suitable for analyzing the transition from conceptual Digital Twin architectures to clinically oriented implementations [1, 2]. This running case also establishes the theoretical basis for the cardiovascular specialization introduced in the PFE.

1.4 Objectives

The main objective of this thesis is to study the state of the art of Digital Twin technology in eHealth and to derive the conceptual, architectural, and evaluation principles required for trustworthy eHealth Digital Twin systems. The specific objectives are:

- to study the evolution, definitions, components, and maturity levels of Digital Twins;
- to review the foundations of eHealth and digital health systems in which Digital Twin principles may be applied;
- to analyze the use of Digital Twins in healthcare from a general eHealth perspective;
- to study major healthcare application domains, including personalized medicine, remote monitoring, hospital processes, public health, and cardiovascular applications;
- to use cardiovascular Digital Twins as a representative clinical deep-dive for multimodal and patient-specific modeling;
- to compare existing architectures, technologies, applications, and limitations;
- to identify major challenges related to interoperability, security, privacy, ethics, and validation;
- to synthesize a multi-layer eHealth Digital Twin architecture and evaluation framework from the reviewed literature;
- to introduce CardioTwin as a practical cardiovascular continuation for the final PFE project.

1.5 Contributions

The main contributions of this thesis are:

- a structured conceptual review of Digital Twin definitions, components, maturity levels, and enabling technologies;
- a classification of healthcare Digital Twin applications from a general perspective;
- a representative cardiovascular deep-dive connecting biosignals, hemodynamics, imaging, physiological modeling, and longitudinal monitoring;
- a comparative study of selected research papers and architectures;
- architecture and evaluation findings derived from the comparative synthesis of the literature;
- a discussion of implementation challenges, research gaps, and future research directions;
- a transition toward the CardioTwin PFE as a practical cardiovascular specialization of the general findings.

1.6 Thesis Outline

This thesis is organized into four main parts. Part I contains the general introduction and presents the research context, problem statement, objectives, and contributions. Part II provides the background: Chapter 2 introduces Digital Twin fundamentals, Chapter 3 presents eHealth and digital-health foundations, and Chapter 4 examines how Digital Twin concepts are applied in healthcare and eHealth, including cardiovascular Digital Twins as a representative clinical deep-dive. Part III presents the state of the art: Chapter 5 defines the structured literature review methodology, while Chapter 6 compares selected studies, synthesizes the main conceptual and architectural findings, derives a multi-layer eHealth Digital Twin architecture, and identifies the remaining research gaps. Part IV contains the final chapter, which concludes the Master study and introduces the proposed PFE method through CardioTwin, a cardiovascular Digital Twin specialization.

Part II

Background

CHAPTER 2

DIGITAL TWIN FUNDAMENTALS

2.1 Origin and Evolution of the Digital Twin Concept

The Digital Twin concept emerged from engineering and product lifecycle management, where virtual representations of physical assets are used to support design, monitoring, prediction, and optimization. Over time, the concept evolved from static digital representations toward dynamic systems connected to real-world data streams and capable of supporting simulation and decision-making [3–5].

In its general form, a Digital Twin can be understood as a virtual representation of a physical entity or process that is connected to the real world through data exchange. The virtual representation is not only a visual model; it may include data, models, algorithms, services, and interfaces used to understand and improve the physical counterpart.

2.2 Digital Model, Digital Shadow, and Digital Twin

One important distinction in the literature is between digital models, digital shadows, and digital twins. A digital model is a virtual representation with no automatic data exchange. A digital shadow includes automatic data flow from the physical entity to the digital representation. A full Digital Twin requires stronger integration, where the physical and virtual entities interact through a bidirectional data flow [4, 6].

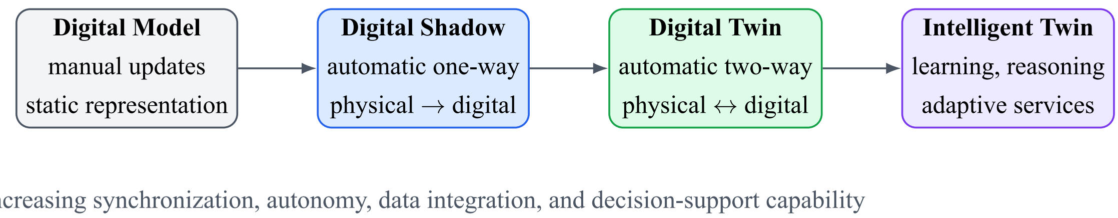

Figure 2.1: Evolution from digital model to intelligent Digital Twin.

2.3 Core Components of a Digital Twin

Most Digital Twin definitions include three fundamental components: a physical entity, a virtual entity, and a connection between them. More extended views also include data and services as explicit components [3, 7]. In this thesis, the Digital Twin is considered as a system composed of five main elements:

1. **Physical space:** the real-world entity, process, device, patient, or healthcare environment.
2. **Virtual space:** the digital representation that models the state and behavior of the physical counterpart.
3. **Data:** real-time, historical, contextual, and simulated data used to feed and update the twin.
4. **Services:** monitoring, simulation, prediction, optimization, alerts, and decision support.
5. **Connection:** communication mechanisms enabling synchronization between the physical and virtual spaces.

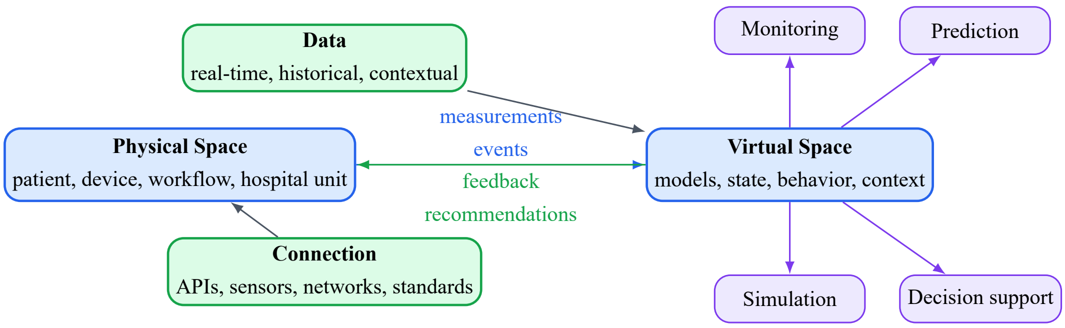

Figure 2.2: Core components of a Digital Twin system.

2.4 Enabling Technologies

Digital Twin systems are enabled by several technologies. The Internet of Things allows physical entities to be monitored through sensors and connected devices. Cloud and edge computing provide infrastructure for data storage and processing. Artificial intelligence and machine learning support pattern recognition, prediction, personalization, and anomaly detection. Simulation models provide interpretable and mechanistic representations of system behavior. Cybersecurity technologies protect data exchange, access control, and system integrity [4, 8, 9].

2.5 Digital Twin Services

The value of a Digital Twin is expressed through its services. Typical services include real-time monitoring, visualization, anomaly detection, risk estimation, scenario simulation, optimization, predictive maintenance, and decision support. In healthcare, these services must be designed carefully because incorrect outputs may affect clinical decisions and patient safety.

2.6 Digital Twin Maturity

Digital Twins can be classified according to their level of maturity. A low-maturity system may only provide a static representation, while a high-maturity system can learn from real-time data, simulate future behavior, interact with the physical world, and support autonomous or semi-autonomous decisions [5, 6]. In healthcare, fully autonomous Digital Twins remain rare because of clinical safety, ethical, legal, and validation requirements.

2.7 Conclusion

This chapter established the conceptual and technological foundations of Digital Twin systems. It clarified the distinction between digital models, digital shadows, Digital Twins, and intelligent Digital Twins, and identified the core role of data, connection, virtual representation, and services. The following chapter introduces the eHealth and digital-health environment in which these principles can be adapted, before their integration in healthcare is examined in the subsequent chapter.

CHAPTER 3

EHEALTH AND DIGITAL HEALTH FOUNDATIONS

3.1 Definition of eHealth

The previous chapter introduced Digital Twin fundamentals, including the distinction between digital models, digital shadows, Digital Twins, and intelligent Digital Twins. This chapter now presents the healthcare environment in which these principles must be adapted. eHealth refers to the use of information and communication technologies to support healthcare services, medical decision-making, prevention, diagnosis, monitoring, treatment, administration, and patient engagement. It includes a wide range of systems such as telemedicine, mobile health, electronic health records, connected medical devices, remote monitoring platforms, and clinical decision support systems.

3.2 From Traditional Healthcare to Connected Healthcare

Traditional healthcare is often centered around episodic consultations, paper-based documentation, and hospital-centered workflows. Connected healthcare extends this model by enabling continuous or near-continuous data acquisition, remote communication, and digital coordination among patients, physicians, caregivers, and healthcare institutions.

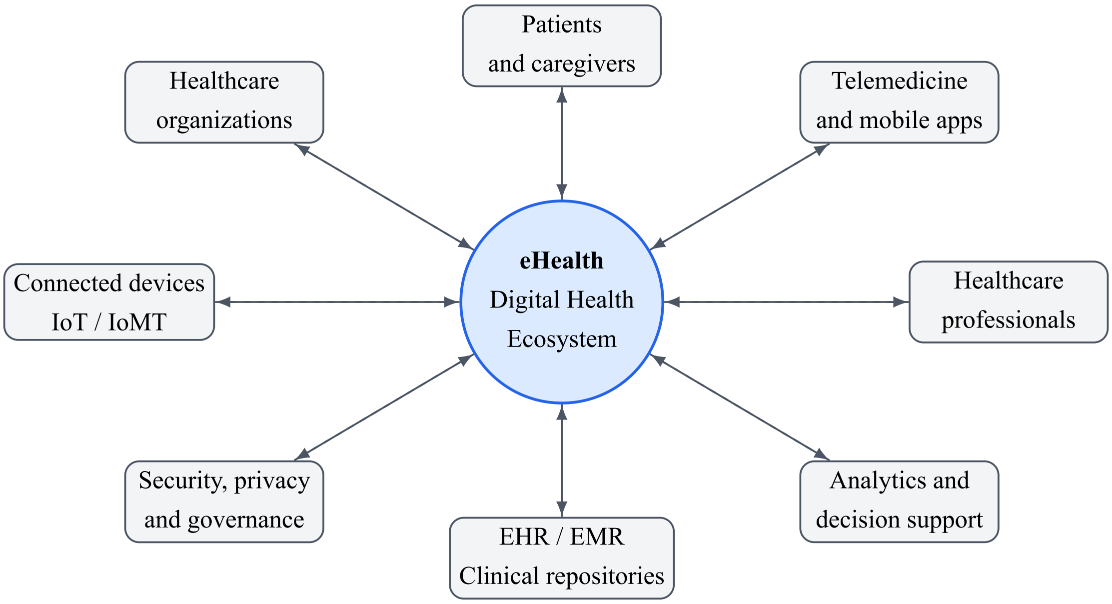

Figure 3.1: General eHealth ecosystem and its main stakeholders.

3.3 Main Components of eHealth Systems

A general eHealth system may include several components:

- **Patient-facing technologies:** mobile applications, wearable sensors, home monitoring devices, and patient portals.
- **Clinical information systems:** EHR, EMR, laboratory information systems, radiology systems, and hospital management platforms.
- **Communication services:** teleconsultation, secure messaging, remote follow-up, and alerts.
- **Analytics and decision support:** machine learning models, rule-based reasoning, risk estimation, dashboards, and report generation.
- **Infrastructure:** cloud platforms, edge devices, APIs, databases, cybersecurity modules, and interoperability standards.

3.4 Data Sources in eHealth

eHealth systems rely on heterogeneous data sources. These include structured clinical data, medical measurements, biosignals, medical images, laboratory results, prescriptions, lifestyle data, environmental data, administrative data, and patient-reported outcomes. The combination of these sources is valuable but challenging due to differences in format, sampling frequency, quality, reliability, and privacy requirements [10, 11].

3.5 Multimodal Data and Longitudinal Follow-Up

eHealth systems increasingly depend on multimodal data. A single health situation may involve structured data from EHRs, biosignals from connected devices, medical images, laboratory results, prescriptions, lifestyle data, and patient-reported outcomes. For Digital Twin development, the temporal dimension is especially important: isolated measurements provide only a snapshot, while longitudinal data make it possible to model trends, detect deterioration, and personalize what is considered normal for a specific patient.

Multimodal integration is difficult because each data source has its own format, sampling frequency, noise level, and clinical meaning. For example, wearable data may be continuous but noisy, imaging data may be detailed but episodic, and EHR data may be rich but incomplete. A robust eHealth architecture must therefore include preprocessing, timestamp alignment, missing data management, and semantic standardization.

3.6 Internet of Medical Things and Edge-Cloud Computing

The Internet of Medical Things (IoMT) connects medical sensors, wearable devices, home monitoring systems, mobile applications, and clinical equipment. These devices can support real-time or near-real-time monitoring, but they also introduce constraints related to energy consumption, connectivity, latency, privacy, and cybersecurity [11]. Edge computing can process sensitive or time-critical data close to the patient, while cloud computing can provide scalable storage, model training, and large-scale analytics.

A Digital Twin-based eHealth system may combine edge and cloud resources. Edge devices can perform first-level filtering, anomaly detection, or local alerts, while cloud services can update patient histories, run heavier simulations, and support long-term analytics. The choice between edge and cloud depends on clinical urgency, bandwidth, model complexity, and privacy requirements.

3.7 Interoperability Standards

Interoperability is essential for eHealth because patient information may be generated and stored by different systems. Standards such as HL7 and FHIR support the exchange of healthcare information using common formats and resources [10, 12, 13]. In a Digital Twin-based eHealth system, interoperability standards help connect data acquisition modules, clinical repositories, virtual models, and decision services.

3.8 Challenges in eHealth

The main challenges of eHealth include data fragmentation, missing data, inconsistent measurements, limited integration between systems, security and privacy risks, lack of clinical validation, usability barriers, and unequal access to digital services. These challenges justify the need for more integrated, dynamic, and trustworthy approaches such as Digital Twin-based architectures.

3.9 Conclusion

This chapter introduced the foundations of eHealth and highlighted the healthcare context in which Digital Twin principles must operate. It showed that eHealth systems depend on heterogeneous data sources, connected devices, interoperability standards, secure infrastructures, and longitudinal patient follow-up. Having introduced both Digital Twin fundamentals and the eHealth ecosystem, the following chapter examines how these two domains are combined in Digital Twin-based healthcare systems.

CHAPTER 4

DIGITAL TWINS IN HEALTHCARE AND EHEALTH

4.1 Digital Twin for Health

The previous chapters introduced the Digital Twin paradigm and then the eHealth environment in which health data, connected devices, clinical information systems, and interoperability standards operate. This chapter connects these two foundations by examining how Digital Twin concepts can be adapted to healthcare and eHealth applications. A Digital Twin for Health is a virtual representation used to support healthcare-related monitoring, simulation, prediction, and decision-making. Depending on its scope, it may represent a patient, an organ, a disease process, a medical device, a clinical workflow, a hospital unit, or a public health system. Unlike a simple dashboard, a Digital Twin aims to maintain a dynamic relationship between real-world data and virtual models.

Recent healthcare literature emphasizes that a health Digital Twin should not be reduced to a static model or a conventional artificial intelligence classifier. It should include a physical entity, a virtual counterpart, and a connection between both, while remaining individualized, interconnected, interactive, informative, and impactful [14]. This definition is important because healthcare applications often use the term Digital Twin for systems that are closer to simulation models, digital shadows, or clinical dashboards. In this thesis, the term is used in a strict but practical sense: an eHealth Digital Twin is a connected virtual representation that can be updated by health data and can provide services such as monitoring, prediction, simulation, and decision support.

4.2 Types of Healthcare Digital Twins

Healthcare Digital Twins can be classified into several categories:

- **Patient-specific Digital Twins:** virtual representations of individual patients using multimodal data.
- **Organ or body-system Digital Twins:** models focusing on a specific organ or physiological system.

- **Disease Digital Twins:** models representing disease progression, risk factors, or treatment response.
- **Medical device Digital Twins:** virtual representations of devices used for design, monitoring, or safety assessment.
- **Hospital and process Digital Twins:** models of workflows, resources, patient flow, logistics, and healthcare units.
- **Public health Digital Twins:** models used for population-level monitoring, planning, prevention, and crisis management.

This taxonomy is consistent with the diversity of healthcare Digital Twin applications reported in the literature. Human Digital Twins may represent the entire human body, a body system, a body organ, a body function, or a disease state, while hospital Digital Twins can represent healthcare processes and resources [14–16]. Figure 4.1 summarizes these categories.

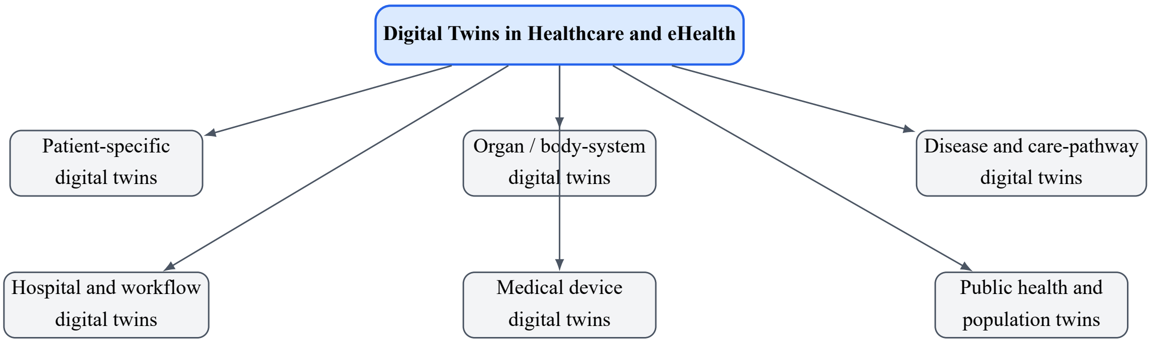

Figure 4.1: General taxonomy of Digital Twin applications in healthcare and eHealth.

4.3 Main Application Domains in Healthcare

Digital Twin applications in healthcare can be organized around several major use cases. Figure 4.2 presents a general map of these application areas.

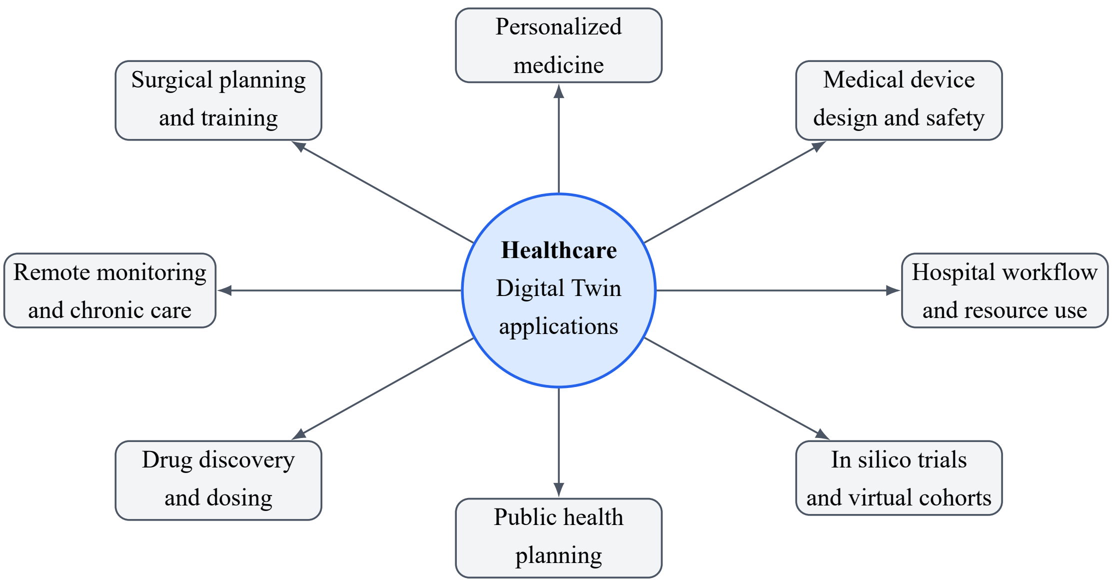

Figure 4.2: Main application domains of Digital Twins in healthcare.

4.3.1 Personalized and Precision Medicine

Personalized medicine is one of the most important motivations for Digital Twins in healthcare. Instead of using only average population models, a patient-specific twin can integrate information about the patient profile, medical history, biological measurements, lifestyle, and response to previous interventions. This makes it possible to support individualized risk assessment, treatment selection, and follow-up [16–18].

In this context, hybrid approaches are particularly important because healthcare decisions often require both data-driven prediction and physiological interpretation. Sharma and Kaur proposed a patient-specific framework that combines mechanistic physiological modeling with deep learning, using heterogeneous clinical data such as EHRs, ECG, imaging, and wearable sensor streams [19]. Such approaches are relevant because purely statistical models can be accurate but difficult to interpret, while purely mechanistic models can be interpretable but hard to personalize at scale.

4.3.2 Remote Monitoring and Chronic Disease Management

Digital Twins can support remote monitoring by continuously or periodically updating a virtual representation from home sensors, wearable devices, mobile applications, and telemedicine platforms. This is relevant for chronic disease management, elderly care, rehabilitation, hospital-at-home programs, and long-term follow-up. A twin can help identify trends, detect anomalies, estimate deterioration risk, and generate alerts for healthcare professionals.

This use case is not limited to one disease. It can be adapted to diabetes, respiratory diseases, cardiovascular diseases, neurological monitoring, mental health, rehabilitation, or frailty monitoring. The main requirement is the availability of reliable longitudinal data and clinically meaningful models.

4.3.3 Hospital Workflow and Resource Optimization

Healthcare Digital Twins can also represent hospitals, departments, or services. In this case, the objective is not only patient physiology but also operational performance. The twin may simulate patient flow, waiting times, bed allocation, staffing, equipment availability, queue management, and emergency department pressure. Noeikham et al. propose a healthcare-unit architecture and show how a Digital Twin can be used for monitoring service systems, predictive analytics, and process optimization in a chemotherapy unit [20].

This operational perspective is important because eHealth is not only about individual monitoring. It also includes the digital transformation of care delivery. Hospital Digital Twins can support administrators and clinicians in testing organizational decisions before applying them to real services, which can reduce risk and improve resource allocation.

4.3.4 Surgical Planning and Clinical Training

Digital Twins may support surgical planning by creating patient-specific anatomical or functional models from imaging and clinical data. These models can help clinicians understand anatomy, test strategies, plan interventions, and train on realistic virtual representations. In surgical contexts, the twin is often closer to an organ-specific or anatomical model, but it can become more dynamic when connected to intraoperative data or postoperative follow-up.

4.3.5 Medical Device Design and Safety

Digital Twins are also used in the development, testing, and monitoring of medical devices. A device Digital Twin may represent the behavior of a device under different physiological conditions, support performance testing, or predict possible failure. In healthcare, this is particularly relevant for implantable devices, prostheses, pacemakers, imaging systems, monitoring devices, and wearable sensors.

4.3.6 Drug Discovery, Dosing, and Treatment Simulation

Another important application is the simulation of pharmacological effects and treatment strategies. Digital Twins can be used to test potential drug responses, optimize dosage, and compare virtual treatment scenarios before making a clinical decision. In practice, this requires reliable biological models, high-quality patient data, and careful clinical validation. The ethical implications are also important because simulated treatment recommendations may influence real patient care.

4.3.7 In Silico Clinical Trials and Virtual Cohorts

Digital Twins can contribute to *in silico* clinical trials by creating virtual patient populations or virtual cohorts. These approaches can be used to test hypotheses, reduce risk, support trial design, and explore treatment effects before or alongside conventional clinical studies [21, 22]. However, such virtual trials must be validated against real clinical evidence and must be transparent regarding assumptions, uncertainty, and bias.

4.3.8 Public Health and Population-Level Digital Twins

At the population level, Digital Twins can support public health monitoring, epidemic response, vaccination planning, resource distribution, and policy simulation. De Benedictis et al. illustrate this direction with CanTwin, a Digital Twin case study for social distancing and public health management in a workplace setting [15]. This shows that Digital Twins in healthcare are not limited to individual patients; they can also support collective health and system-level decision-making.

4.4 Patient-Centered eHealth Digital Twins

A patient-centered eHealth Digital Twin integrates clinical, physiological, behavioral, and contextual data to support personalized monitoring and decision-making. Such a system may receive information from EHRs, connected devices, laboratory results, patient-reported outcomes, and environmental sources. However, the level of clinical reliability depends on data quality, model validation, and the ability to handle uncertainty.

Patient-centered Digital Twins require a data lifecycle that includes acquisition, preprocessing, alignment, storage, modeling, simulation, service delivery, and feedback. They also require an explicit distinction between observed data and inferred state variables. For example, a sensor may directly measure heart rate or activity level, while a model may estimate risk, disease progression, treatment response, or latent physiological parameters. The inferred outputs should therefore be presented with uncertainty and should not be confused with direct measurements.

4.5 Healthcare Process Digital Twins

Digital Twins are not limited to individual patients. They can also represent healthcare processes, such as emergency department flow, outpatient services, operating room scheduling, hospital bed allocation, or vaccination campaigns. In this context, the Digital Twin supports simulation of operational scenarios and optimization of healthcare resources.

A process Digital Twin generally requires event data, timestamps, resource availability, staff schedules, patient pathways, and performance indicators. Its objective is to support operational decisions such as allocating staff, reducing waiting time, improving patient flow, detecting bottlenecks, and planning resources under uncertainty. This type of twin is important for eHealth because digital health services must be integrated into real healthcare organizations, not only into individual mobile applications.

4.6 Cardiovascular Digital Twins as a Representative Clinical Deep-Dive

Although this thesis studies eHealth in a general sense, the cardiovascular field is used as a representative clinical deep-dive. Cardiovascular diseases are complex, dynamic, and heterogeneous. They involve electrical, mechanical, hemodynamic, anatomical, behavioral, and clinical factors. For this reason, cardiovascular Digital Twins provide a useful example of how eHealth Digital Twins can combine multimodal data, mechanistic modeling, artificial intelligence, and clinical decision support [1, 2].

4.6.1 Why Cardiovascular Digital Twins?

Cardiovascular care often requires individualized interpretation of signals and measurements such as ECG, blood pressure, imaging, cardiac volumes, ejection fraction, pulmonary pressures, medication history, comorbidities, symptoms, and laboratory tests. Coorey et al. describe the health Digital Twin for cardiovascular disease as an emerging interdisciplinary field in which real-time patient data and virtual representations can be used for prediction and treatment optimization [1]. This makes cardiovascular care a strong candidate for Digital Twin development because the same disease label may correspond to different underlying mechanisms in different patients.

4.6.2 Data Sources for Cardiovascular Digital Twins

A cardiovascular Digital Twin can integrate several data categories:

- **Biosignals:** ECG, photoplethysmography, heart rate variability, and wearable sensor streams.
- **Hemodynamic measurements:** systolic and diastolic blood pressure, pulmonary artery pressure, cardiac output, and vascular resistance.
- **Medical imaging:** echocardiography, CT, MRI, angiography, and anatomical segmentation.
- **Clinical data:** EHR, diagnosis, symptoms, medication, laboratory tests, and comorbidities.
- **Lifestyle and contextual data:** activity, sleep, diet, stress, environment, and adherence information.

These heterogeneous sources are useful because a Digital Twin is expected to represent more than a single measurement. However, they also increase the difficulty of data alignment, missing data management, validation, and privacy protection.

4.6.3 Organ-Level Heart and Hemodynamic Twins

Organ-level cardiovascular twins aim to reproduce the structure or function of the heart, vessels, or circulatory system. They can be used to simulate blood flow, cardiac mechanics, electrophysiology, valve function, device behavior, or hemodynamic response. In precision cardiology, such models may support noninvasive evaluation of physiology, testing of interventions, and better understanding of patient-specific mechanisms [2].

Geddes et al. present a proof-of-concept framework for heart failure Digital Twins that uses CT pulmonary angiography and hemodynamic data to model pulmonary arterial pressure using computational fluid dynamics [23]. The study is important because it shows how Digital Twins can estimate clinically relevant variables that are otherwise difficult or invasive to measure continuously.

4.6.4 Heart Failure Digital Twins

Heart failure is a particularly relevant use case because it is heterogeneous and difficult to personalize using simple clinical variables alone. Gu et al. developed mechanistic computational models of the cardiovascular system for 343 heart failure patients and used Digital Twin-derived features to guide interpretable AI for diagnosis and prognosis [24]. Their work shows that Digital Twin features can help identify phenogroups, mechanistic drivers, and prognostic indicators that may be more interpretable than raw clinical variables alone.

This example illustrates a central advantage of Digital Twins in healthcare: they can transform heterogeneous measurements into structured physiological features. These features can then be used by AI models to improve prediction while maintaining a link to physiological meaning.

4.6.5 Electrophysiology and Arrhythmia Applications

Another cardiovascular direction is electrophysiology. Patient-specific cardiac electrical models can help simulate activation patterns, arrhythmia mechanisms, and response to interventions such as ablation or cardiac resynchronization therapy. These applications often require ECG, imaging, anatomical segmentation, and electrophysiological constraints. They demonstrate how a Digital Twin can serve not only as a monitoring tool, but also as a simulation environment for therapy planning.

4.6.6 Cardiovascular Digital Twin Data Flow

Figure 4.3 summarizes a general cardiovascular Digital Twin data flow. The same pattern can be adapted to other body systems by changing the data sources and domain-specific models.

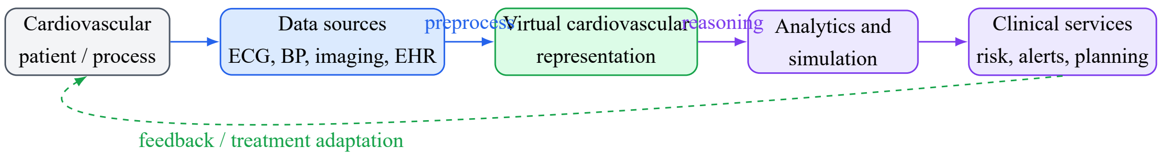

Figure 4.3: General data flow for cardiovascular Digital Twin applications.

4.6.7 Benefits and Limitations in Cardiovascular Applications

Cardiovascular Digital Twins can improve understanding of disease heterogeneity, support personalized risk estimation, enable virtual testing of interventions, and provide interpretable physiological features for AI models. However, they also face important limitations. High-quality multimodal data may be difficult to obtain, imaging and invasive measurements are not always available, models may require calibration, and predictions must be validated before clinical use. Furthermore, real-time synchronization is still challenging because high-fidelity physiological simulations can be computationally expensive.

4.7 Ethical, Legal, and Social Considerations

Healthcare Digital Twins use sensitive data and may influence medical decisions. Therefore, their development must consider consent, privacy, data ownership, transparency, fairness, bias mitigation, accountability, and clinical responsibility. Ethical design is not an optional component; it is a necessary condition for trustworthy eHealth Digital Twins.

Bruynseels et al. emphasize that health Digital Twins may redefine the meaning of health, normality, prevention, and enhancement because they can compare an individual against both personal longitudinal patterns and population-level patterns [25]. This creates opportunities for earlier and more personalized care, but it also raises risks of discrimination, unequal access, overmedicalization, and misuse of sensitive data. Related AI-in-healthcare literature also highlights the need for accountability, fairness, transparency, and explainability when algorithmic outputs may influence clinical decisions [26–29].

4.8 Challenges and Limitations

The implementation of Digital Twins in eHealth faces several challenges: interoperability between healthcare systems, access to high-quality multimodal data, real-time synchronization, computational cost, cybersecurity, explainability, regulatory approval, integration into clinical workflows, and acceptance by healthcare professionals and patients.

The literature also highlights additional issues: data bias, lack of representative datasets, missing longitudinal data, validation difficulties, unclear responsibility when recommendations are wrong, and the gap between academic prototypes and clinically deployable systems [14, 16]. For cardiovascular applications, these challenges are especially visible because models often depend on specialized imaging, invasive measurements, or complex simulations.

4.9 Conclusion

Digital Twins offer a promising direction for eHealth, but their adoption requires more than technical models. It requires architecture, governance, validation, security, interoperability, and ethical alignment. The cardiovascular field illustrates how Digital Twins can move from general monitoring to advanced multimodal modeling and interpretable decision support. These issues are addressed in the state-of-the-art chapters that follow.

Part III

State of the Art

CHAPTER 5 LITERATURE REVIEW METHODOLOGY

5.1 Review Objective

The objective of this chapter is to define the method used to build the state-of-the-art analysis of Digital Twins for eHealth. The review is structured and focused, but it is not presented as an exhaustive protocol-based evidence synthesis. No complete reproducible search log with database-specific dates, initial record counts, duplicate-removal counts, screening counts, full-text exclusion counts, and exclusion reasons is available in this project. For this reason, the methodology is reported as a structured state-of-the-art literature review.

The review aims to connect several bodies of literature: general Digital Twin foundations, healthcare and eHealth applications, healthcare-oriented architectures, patient-specific modeling, cardiovascular examples, interoperability, privacy, security, ethics, explainability, and validation. The purpose is not to cover every publication using the term Digital Twin, but to select high-relevance peer-reviewed papers and authoritative sources that support the conceptual and architectural synthesis developed in Chapter 6.

5.2 Research Questions

The review is guided by the following research questions:

- **RQ1:** What are the main definitions, components, and maturity levels of Digital Twin technology?
- **RQ2:** How is Digital Twin technology applied in healthcare and eHealth?
- **RQ3:** What architectures have been proposed for healthcare Digital Twins?
- **RQ4:** What technologies enable Digital Twin-based eHealth systems?
- **RQ5:** What are the main challenges and limitations of Digital Twins in eHealth?
- **RQ6:** How is Digital Twin technology used in the cardiovascular field, and what can this domain teach us about patient-specific and clinically interpretable eHealth systems?

5.3 Scope of the Review

The review covers two complementary levels. The first level is general and concerns Digital Twin concepts, enabling technologies, healthcare applications, architectures, and trustworthy deployment requirements. This level supports the general eHealth orientation of the thesis. The second level is a representative clinical deep-dive into cardiovascular Digital Twins. This domain is used because it combines biosignals, imaging, hemodynamic measurements, longitudinal monitoring, patient heterogeneity, and patient-specific physiological modeling [1, 2, 23, 24].

The review therefore includes both broad review papers and selected original studies. Broad reviews are used to clarify definitions, taxonomy, architecture, and recurring challenges [4, 5, 14]. Original or application-oriented studies are used to illustrate concrete architectures, healthcare-unit twins, patient-specific frameworks, and cardiovascular modeling approaches [15, 19, 20, 23, 24].

5.4 Source Identification Strategy

Sources were identified through focused searches and citation tracing from high-relevance review papers. The platforms and sources consulted included publisher pages and indexing platforms such as IEEE Xplore, PubMed/MEDLINE, ScienceDirect, Springer Nature, MDPI, official DOI pages, and official standards pages when relevant. These platforms are reported as source-identification channels, not as evidence that a database-by-database protocol search with final numerical counts was performed.

The search concepts combined Digital Twin terminology with healthcare, architecture, interoperability, validation, ethics, and cardiovascular terms. Examples include:

- "digital twin" AND healthcare;
- "digital twin" AND eHealth;
- "digital twin for health";
- "patient-specific digital twin";
- "healthcare digital twin architecture";
- "cardiovascular digital twin";
- "heart failure digital twin";
- "digital twin" AND interoperability;
- "digital twin" AND ethics AND healthcare;
- "digital twin" AND clinical validation.

Additional sources were selected from the reference lists of key review papers when the original paper could be verified and when its contribution was relevant to the comparative analysis. The PFE topic was used only as a practical motivation for the cardiovascular continuation; it was not cited as an external scientific authority.

5.5 Selection Criteria

5.5.1 Inclusion Criteria

Studies were included when they satisfied at least one of the following criteria:

- they define or clarify the Digital Twin concept;
- they distinguish digital models, digital shadows, and Digital Twins;
- they propose healthcare-oriented Digital Twin architectures;
- they present healthcare Digital Twin applications;
- they examine patient-specific modeling;
- they discuss AI and simulation integration;
- they present cardiovascular or heart-failure applications;
- they address interoperability, privacy, ethics, validation, explainability, or governance;
- they are sufficiently detailed to support the comparative analysis.

5.5.2 Exclusion Criteria

Studies were excluded when they:

- used the term Digital Twin without a clear physical–virtual relationship;
- were unrelated to healthcare or to transferable Digital Twin foundations;
- provided no meaningful conceptual, architectural, application, or evaluation contribution;
- were duplicates of already selected sources;
- were inaccessible in sufficient detail;
- were purely promotional or unsupported commercial material;
- confused a simple simulation or dashboard with a mature Digital Twin without justification.

5.6 Data Extraction and Classification

For each selected source, the review extracted the following information when available:

- publication type and scope;
- physical entity, patient, organ, device, or healthcare process represented;
- data modalities and data sources;

- modeling approach, including mechanistic simulation, AI, hybrid modeling, or rule-based reasoning;
- synchronization direction and maturity level;
- services provided, such as monitoring, prediction, simulation, treatment planning, workflow optimization, or decision support;
- validation or evidence reported by the authors;
- interoperability, security, privacy, ethics, explainability, and governance considerations;
- limitations and future research directions.

The extracted information was then classified into thematic groups: Digital Twin foundations, Digital Twins for Health, healthcare architectures, patient-specific and hybrid frameworks, cardiovascular Digital Twins, and interoperability. Ethics, governance, explainability, and validation were treated as cross-cutting themes.

5.7 Synthesis Method

The synthesis was qualitative and comparative. The selected studies were compared across recurring dimensions rather than summarized one by one. Particular attention was given to conceptual maturity, physical–virtual coupling, data modalities, synchronization, modeling strategy, clinical or operational service, validation level, interoperability, explainability, and governance.

Figure 5.1 summarizes the structured workflow used in this review.

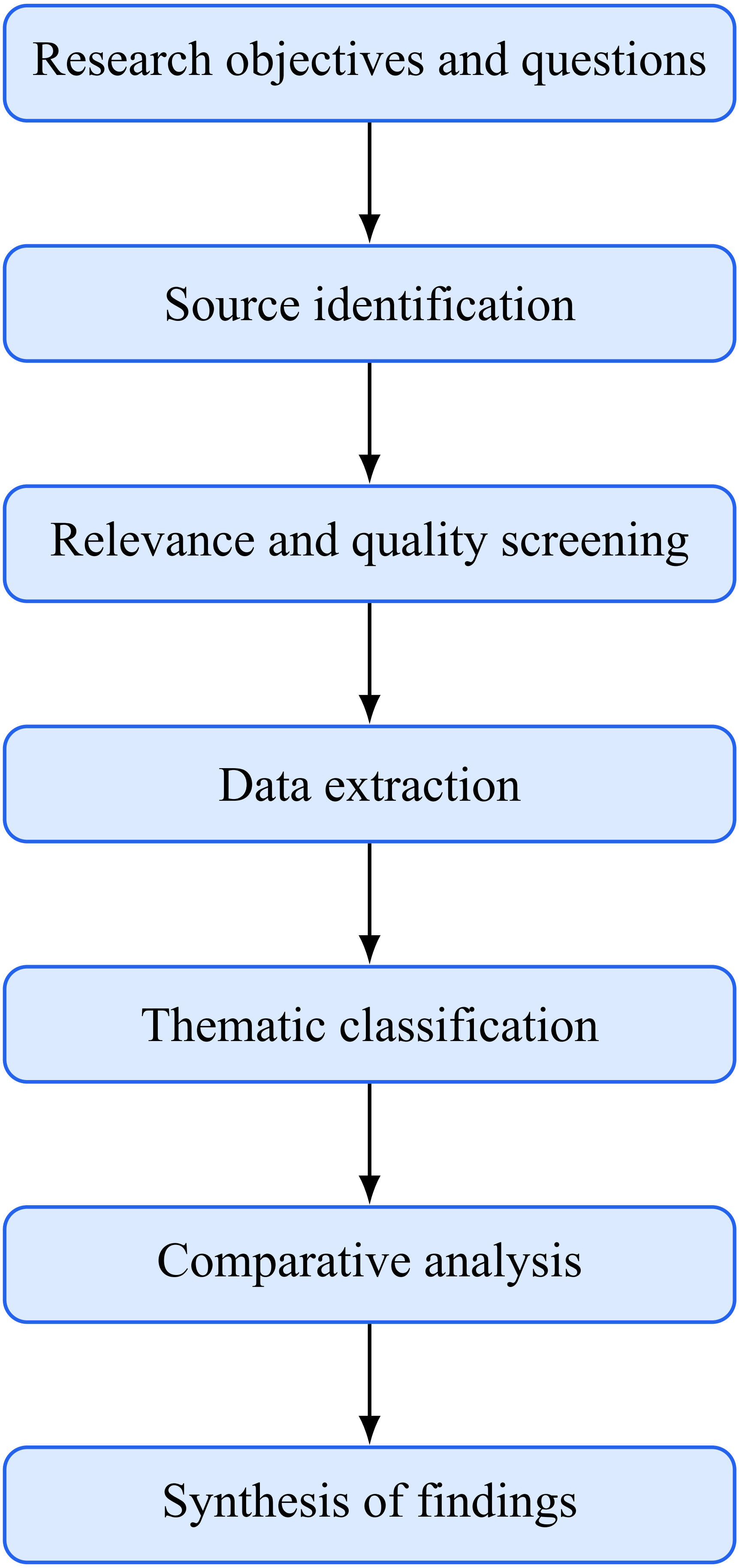

Figure 5.1: Workflow of the structured state-of-the-art literature review.

The synthesis supports the comparative table and the findings developed in Chapter 6. It is designed to identify recurring architectural components, cross-cutting requirements, methodological limitations, and research gaps rather than to calculate a meta-analytic effect size or produce statistical evidence.

5.8 Methodological Limitations

The main limitation of this review is that it is not exhaustive. It does not report database-specific search dates, exact record counts, duplicate-removal counts, exclusion counts, or a complete search log. As a result, it should be interpreted as a structured and focused state-of-the-art review rather than an exhaustive evidence synthesis.

Another limitation is the heterogeneity of Digital Twin terminology. Some publications use the term Digital Twin for static models, offline simulations, dashboards, or digital shadows. The review therefore interprets maturity cautiously and avoids treating all reported systems as fully synchronized and clinically validated Digital Twins. Finally, the field is evolving rapidly, so the reviewed literature represents a selected and verified evidence base rather than a final boundary of the domain.

5.9 Conclusion

This chapter defined the structured methodology used to analyze Digital Twin research for eHealth. It clarified the review objective, research questions, scope, source-identification strategy, selection criteria, data-extraction dimensions, synthesis method, and methodological limitations. The next chapter applies this framework to compare selected studies and derive the main conceptual, architectural, clinical, and evaluation findings.

CHAPTER 6

COMPARATIVE STUDY OF DIGITAL TWIN APPROACHES IN EHEALTH

6.1 Overview of Selected Studies

The selected studies are organized into seven categories: Digital Twin foundations, Digital Twins for Health, healthcare architecture proposals, patient-specific and hybrid AI–simulation frameworks, interoperability sources, ethics and governance sources, and cardiovascular Digital Twin studies. This classification supports comparison rather than isolated paper summaries.

General Digital Twin papers clarify definitions, maturity levels, physical–virtual coupling, enabling technologies, and architecture patterns [3–6, 8]. Healthcare-oriented reviews and perspectives explain adaptation to patient-specific modeling, precision medicine, clinical workflows, and public health [14, 16–18, 22]. Architecture-oriented studies support comparison between healthcare-unit, process-oriented, and patient-oriented designs [15, 19, 20].

Cardiovascular studies are included as a representative clinical deep-dive because they combine biosignals, imaging, hemodynamics, physiological simulation, and AI in patient-specific models [1, 2, 23, 24]. Ethical, interoperability, and validation sources are also included because trustworthy Digital Twins require responsible data use, explainability, standards-based integration, clinical validation, and human supervision [10, 12, 21, 25, 26, 29].

6.2 Analytical Comparison Dimensions

The comparison uses the following dimensions: represented entity or process, data modalities, modeling approach, synchronization direction, maturity level, services, evidence or validation strategy, interoperability, explainability, privacy, governance, and limitations. These dimensions distinguish conceptual frameworks, static models, digital shadows, partial Digital Twins, process Digital Twins, patient-specific calibrated models, and intelligent-DT visions.

This distinction is necessary because Digital Twin terminology remains inconsistent. Maturity labels are therefore used cautiously: a system is not treated as a complete real-time clinical Digital Twin unless it demonstrates continuous data exchange, synchronization with an active physical counterpart, virtual-state update, and clinically meaningful service integration.

6.3 Comparative Table

Table 6.1 compares selected studies across conceptual, architectural, technical, clinical, and governance dimensions. The table is placed in landscape orientation to preserve readability across pages.

Table 6.1: Comparative overview of selected Digital Twin studies for eHealth.

Study	Scope	Entity / process	Data modalities	Modeling approach	Maturity	Services / evidence	Main limitations
Kritzinger et al. [6]	DT classification	Industrial assets and production systems	Production and lifecycle data	Conceptual classification	Maturity taxonomy	Distinguishes digital model, digital shadow, and Digital Twin.	Manufacturing focus; healthcare adaptation requires clinical validation and governance.
Fuller et al. [4]	General DT review	Multi-domain physical entities	Sensor, operational, historical, and simulation data	Review of enabling technologies	Conceptual foundation	Clarifies definitions, misconceptions, technologies, applications, and open challenges.	Not healthcare-specific; clinical safety and privacy are not the central focus.
Tao et al. [3]	Industrial DT architecture	Industrial products and systems	Physical, virtual, service, and connection data	Five-dimensional DT model	Architecture model	Supports physical–virtual coupling, services, and data-driven operation.	Industrial orientation; patient-specific clinical constraints are not addressed.
Jones et al. [5]	DT characterization	Multi-domain entities	Literature-derived dimensions	Conceptual characterization	Definition synthesis	Identifies gaps around lifecycle use, fidelity, data ownership, and integration.	Broad scope; does not validate a healthcare implementation.
Rasheed et al. [8]	Modeling perspective	Complex physical systems	Simulation, sensor, and computational-model data	Modeling and simulation review	Modeling foundation	Discusses values, challenges, and enablers for adaptive model-based twins.	General modeling focus; clinical workflow integration is outside scope.
Katsoulakis et al. [14]	DT for Health review	Patients, organs, workflows, public health systems	Multimodal clinical and digital-health data	Scoping review	Field-level synthesis	Maps healthcare applications, challenges, enabling technologies, and opportunities.	Demonstrates that the field remains heterogeneous and often conceptual.
Björnsson et al. [17]	Personalized medicine	Individual patients	Molecular, clinical, and patient-specific data	High-resolution patient models	Precision-medicine vision	Proposes Digital Twins for individualized drug testing and treatment selection.	Conceptual and translational; requires data integration and clinical validation.

Continued on next page

Table 6.1: Comparative overview of selected Digital Twin studies for eHealth (continued).

Study	Scope	Entity / process	Data modalities	Modeling approach	Maturity	Services / evidence	Main limitations
Venkatesh et al. [16]	Health DT implementation	Patient and population health twins	Multimodal patient, population, and real-time data	Perspective on implementation	Health-DT vision	Discusses computation, clinical implementation, governance, and regulation.	Does not provide a deployed clinical twin; emphasizes implementation barriers.
De Benedictis et al. [15]	Healthcare architecture	Healthcare processes and public-health case	Organizational, environmental, and process data	Six-layer architecture	Process DT / architecture	Proposes layered architecture and applies it to a social-distancing case study.	Process-oriented; patient-specific physiology and organ-level models need extension.
Noeikham et al. [20]	Healthcare-unit architecture	Chemotherapy service unit	EHR/CDR and healthcare-service data	Five-layer healthcare-unit architecture	Process DT	Emphasizes healthcare ICT integration, security, privacy, and service monitoring.	Focused on a unit-level scenario; broader eHealth scalability remains to be evaluated.
Sharma and Kaur [19]	Patient-specific framework	Individual patient state	EHR, ECG, imaging, wearables, and clinical data	Hybrid AI and physiological simulation	Intelligent-DT framework	Integrates multimodal data, deep learning, simulation, and personalization.	Requires external validation, clinical workflow evaluation, and regulatory analysis.
Lehne et al. [10]	Digital medicine interoperability	Health IT systems	EHR and multi-institutional health data	Perspective on interoperability	Enabling infrastructure	Argues that digital medicine depends on technical and semantic interoperability.	Not a Digital Twin paper; supports the data-integration requirement.
Mandel et al. [12]	SMART on FHIR	EHR application ecosystem	FHIR resources and EHR app data	Standards-based app platform	Interoperability enabler	Shows how standards-based apps can run across EHR environments.	Not a DT implementation; relevant as an integration mechanism.
Bruynseels et al. [25]	Ethics and governance	Human and health twins	Sensitive personal and health data	Ethical analysis	Governance foundation	Discusses privacy, discrimination, identity, health meaning, and responsibility.	Conceptual ethics focus; no technical architecture or clinical validation.
Amann et al. [29]	Explainable medical AI	AI-based clinical decision support	Clinical decision-support data	Multidisciplinary explainability analysis	Governance / usability enabler	Shows why explainability has technical, legal, medical, and patient dimensions.	Not DT-specific; supports explainability requirements for DT services.
Corral-Acero et al. [2]	Precision cardiology	Cardiovascular patient models	Imaging, electrophysiology, clinical and mechanistic data	Statistical and mechanistic modeling	Precision-cardiology vision	Describes the cardiovascular Digital Twin as a route toward personalized cardiology.	Mainly position and review; practical deployment and validation remain challenging.

Continued on next page

Table 6.1: Comparative overview of selected Digital Twin studies for eHealth (continued).

Study	Scope	Entity / process	Data modalities	Modeling approach	Maturity	Services / evidence	Main limitations
Coorey et al. [1]	Cardiovascular DT review	Cardiovascular disease and care pathways	Biosignals, imaging, clinical records, devices, patents	Review of CVD DT field	Emerging health-DT field	Maps cardiovascular DT concepts, research, industry activity, and challenges.	Many CVD examples are precursor models rather than synchronized clinical twins.
Martinez-Velazquez et al. [30]	Edge heart twin	Human heart representation	Physiological and edge-computing data	Edge-based heart-twin prototype	Prototype / partial DT	Demonstrates an edge-oriented Cardio Twin concept for heart modeling.	Prototype scale; limited clinical validation and patient-specific evidence.
Geddes et al. [23]	Heart-failure monitoring	Pulmonary arteries and right-heart failure markers	CT pulmonary angiography and hemodynamic data	Computational fluid dynamics	Patient-specific calibrated model	Estimates pulmonary artery pressure and right-heart failure markers noninvasively.	Proof-of-concept; depends on imaging, boundary conditions, computation, and validation.
Gu et al. [24]	Heart-failure diagnosis and prognosis	Cardiovascular system models for HF patients	Clinical variables and DT-derived physiological features	Mechanistic cardiovascular model plus interpretable AI	Patient-specific model / intelligent-DT direction	Uses DT-derived features to support diagnosis, phenotyping, and prognosis in 343 HF patients.	Requires multicenter prospective validation and broader longitudinal testing.

6.4 Synthesis of Findings

6.4.1 Conceptual Findings

The comparative analysis shows that Digital Twin terminology remains inconsistent. Foundational papers converge on the need for a physical entity, a virtual representation, and a meaningful connection between them, but healthcare literature often applies the term to systems with different levels of maturity [4–6]. Some studies describe static anatomical models, offline simulations, monitoring dashboards, or one-way data pipelines as Digital Twins. These systems may be valuable, but they should be distinguished from fully synchronized twins.

The distinction between a digital model, a digital shadow, a Digital Twin, and an intelligent Digital Twin is therefore central to the thesis. A true Digital Twin requires more than a simulation model; it requires a maintained relationship between the physical and virtual entities. Data and services are also important dimensions because the value of a twin depends on the ability to update, interpret, simulate, and expose useful outputs to users [3, 7]. In healthcare, maturity must also account for clinical safety, validation, interpretability, and workflow integration.

6.4.2 Data Findings

Healthcare Digital Twins require multimodal and longitudinal data. Patient-specific systems may combine EHR data, biosignals, imaging, laboratory values, medication history, wearable measurements, patient-reported information, and contextual variables [14, 16, 19]. The comparative literature shows that the main difficulty is not only collecting data, but aligning them temporally, validating their quality, and preserving their clinical meaning.

Measurements may be incomplete, contradictory, noisy, or collected at different sampling rates. Wearable and IoMT data can support continuous monitoring, but they introduce reliability, calibration, connectivity, and privacy concerns [11]. Interoperability remains a central limitation because clinical data are often fragmented across EHR systems, repositories, devices, and organizational boundaries [10, 12]. Data provenance and quality must therefore be treated as architectural requirements, not as preprocessing details.

6.4.3 Modeling Findings

The reviewed studies show that mechanistic models and AI models provide different but complementary strengths. Mechanistic models improve physiological interpretability and can support what-if simulation, especially in domains such as cardiovascular mechanics, electrophysiology, and hemodynamics [2, 23]. AI methods support pattern recognition, risk prediction, phenotyping, anomaly detection, and adaptation to complex multimodal data [19, 24].

Hybrid approaches are promising because they combine data-driven prediction with physiological constraints derived from domain knowledge [8, 18]. However, both approaches have limitations. Mechanistic models may be computationally expensive and difficult to calibrate for individual patients. AI models may be sensitive to bias, missing data, and poor generalization. Uncertainty must therefore be represented and communicated, especially when model outputs may influence clinical interpretation.

6.4.4 Architectural Findings

Architecture-oriented studies converge toward layered and modular healthcare Digital Twin designs. De Benedictis et al. propose a generalized healthcare architecture, while Noeikham et al. emphasize healthcare-unit integration, ICT systems, security, privacy, and service monitoring [15, 20]. Broader Digital Twin architecture literature also separates the physical space, data acquisition, data management, virtual representation, analytics, service delivery, and connection mechanisms [3, 4].

The synthesis of these studies indicates that a healthcare Digital Twin architecture should not be a monolithic system. Patient-specific twins and healthcare-unit twins have different modeling requirements, but both need data acquisition, interoperability, validated storage, virtual-state management, intelligence or simulation components, and user-facing services. Security, privacy, governance, explainability, traceability, and consent management must operate across all layers rather than being added as a final component.

6.4.5 Derived Multi-Layer Architecture

From the reviewed literature, seven recurring architectural layers can be synthesized for a general eHealth Digital Twin system:

- **Physical Layer**: represents the real healthcare environment, including patients, medical devices, sensors, clinicians, hospital units, and clinical processes.

- **Data Acquisition Layer**: collects data from connected sensors, electronic health records, laboratory results, imaging systems, wearable devices, and clinical inputs.

- **Interoperability and Preprocessing Layer**: cleans, validates, aligns, standardizes, and transforms heterogeneous data so that they can be used by the twin.

- **Data Management Layer**: stores structured and unstructured data, maintains longitudinal records, manages metadata, and supports secure access to patient-related information.

- **Virtual Twin Layer**: maintains the virtual representation of the patient, organ, device, process, or healthcare service being modeled.

- **Intelligence and Simulation Layer**: applies AI, analytics, mechanistic simulation, risk estimation, and what-if analysis to support interpretation and prediction.

- **Service Layer**: provides monitoring, visualization, alerts, reports, decision support, and controlled feedback to healthcare professionals, patients, or administrators.

Figure 6.1 summarizes this multi-layer architecture as a result of the state-of-the-art synthesis.

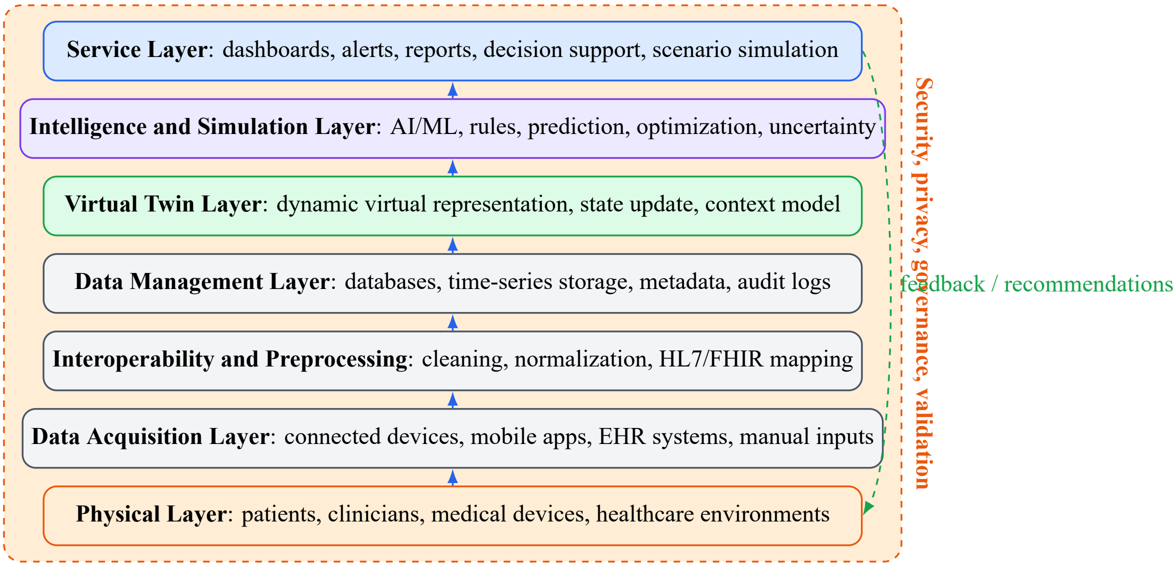

Figure 6.1: Multi-layer eHealth Digital Twin architecture derived from the state-of-the-art synthesis.

The figure should be interpreted as a general research synthesis rather than as a final implementation design. It identifies recurring layers and clarifies how data, virtual representation, intelligence, simulation, and services interact. The lower layers connect the real healthcare environment to validated and interoperable data. The middle layers preserve longitudinal context and update the virtual representation. The upper layers transform the twin into monitoring, prediction, simulation, and decision-support services.

6.4.6 Cross-Cutting Requirements

The literature shows that several requirements must affect every layer. Security, privacy, governance, interoperability, ethics, traceability, consent management, explainability, clinical validation, and human supervision are cross-cutting conditions for trustworthy deployment [25–27, 29]. For example, privacy begins during acquisition, continues through storage and modeling, and remains relevant when outputs are displayed or reused. Explainability depends not only on AI models but also on data provenance, model assumptions, uncertainty communication, and clinical interpretation.

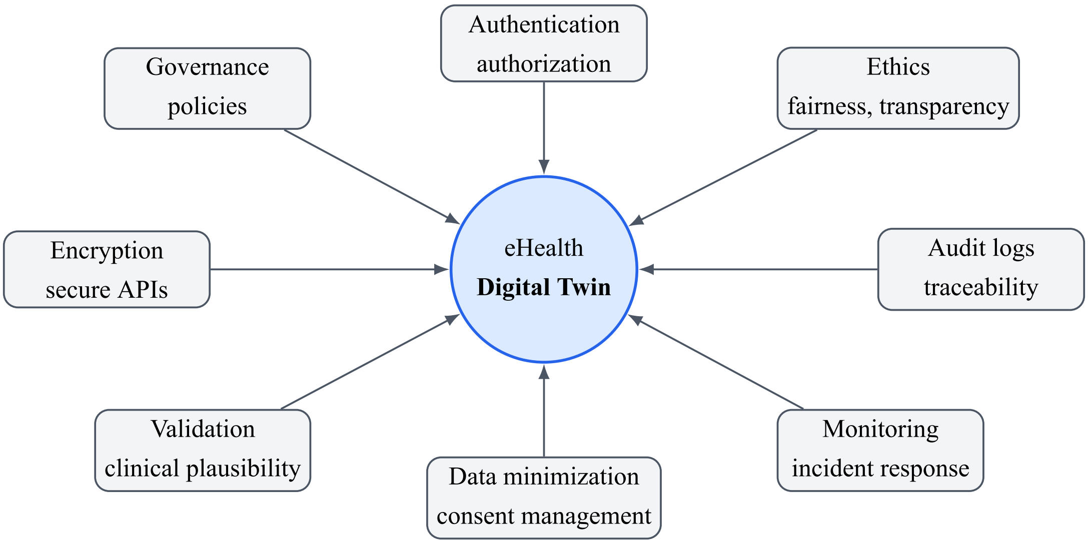

Figure 6.2: Cross-cutting security, privacy, governance, and ethical requirements synthesized for eHealth Digital Twin systems.

6.4.7 Data Flow and Service Logic

The synthesized data-flow logic follows a progressive sequence: the physical healthcare environment generates biomedical, clinical, behavioral, or organizational data; the data acquisition layer captures these data; preprocessing and interoperability mechanisms validate, clean, normalize, align, and standardize them; the data management layer stores them with longitudinal context; the virtual twin is updated; AI and simulation components analyze the updated state; services present outputs to users; and feedback to the real environment remains controlled and supervised.

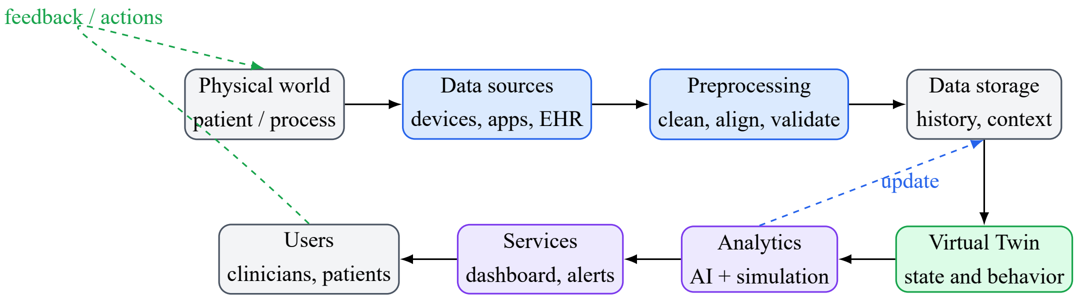

Figure 6.3: Synthesized data and feedback flow for a Digital Twin-based eHealth system.

This workflow emphasizes that clinical feedback should normally remain human-supervised and validated. In clinical environments, recommendations, alerts, treatment scenarios, or simulated outcomes must be interpreted by healthcare professionals and integrated with medical judgment, patient context, and ethical responsibility.

6.4.8 Clinical and Cardiovascular Findings

Most healthcare Digital Twin systems remain conceptual, retrospective, simulated, or proof-of-concept. Prospective clinical validation is still uncommon, and bidirectional autonomous feedback is rare. This does not reduce the value of the research, but it requires cautious terminology. A system that predicts risk from retrospective data is not equivalent to a clinically deployed and continuously synchronized patient twin.

Cardiovascular Digital Twins demonstrate the value of combining biosignals, imaging, pressure or flow measurements, and physiological models. The cardiovascular literature includes broad field mapping, precision-cardiology visions, edge prototypes, patient-specific hemodynamic simulations, and heart-failure phenotyping using Digital Twin-derived features [1, 2, 23, 24, 30]. These studies show how hybrid modeling can transform heterogeneous measurements into clinically interpretable features. At the same time, many cardiovascular works are patient-specific simulations or calibrated models rather than fully synchronized clinical Digital Twins.

6.4.9 Evaluation Requirements

Evaluation must cover several dimensions. Technical validation concerns functional completeness, interoperability, synchronization quality, latency, scalability, reliability, security, privacy, auditability, and robustness. Model validation concerns predictive performance, physiological plausibility, uncertainty communication, explainability, calibration, and sensitivity to missing or noisy data. Clinical validation concerns safety, clinical plausibility, usefulness, prospective evaluation, comparison with accepted clinical practice, and clinician review. Workflow validation concerns usability, dashboard clarity, integration into routine practice, alert fatigue, user trust, and the effect of the system on care processes [21, 29].

These findings contribute directly to the general eHealth architecture by defining the layers, data-flow logic, and cross-cutting requirements that any trustworthy eHealth Digital Twin should address. They also prepare the PFE transition by showing why a cardiovascular prototype should focus on multimodal data acquisition, patient-state representation, AI-assisted ECG interpretation, controlled simulation, explainability, and human-in-the-loop decision support.

6.5 Research Gap

The synthesis identifies several unresolved problems. First, there is still no universally accepted definition of healthcare Digital Twins, and maturity classification remains inconsistent. Many systems use the term Digital Twin without demonstrating bidirectional synchronization, continuous state update, or clinically integrated services.

Second, healthcare data remain fragmented, heterogeneous, and difficult to integrate temporally. Interoperability across EHRs, devices, imaging systems, clinical repositories, and patient-generated data remains limited. Missing values, contradictory measurements, biased datasets, weak provenance, and insufficient longitudinal context can reduce the reliability of patient-state representation.

Third, validation remains a major gap. Many works are evaluated retrospectively, conceptually, or on limited proof-of-concept data. External validation, prospective evaluation, clinician review, workflow testing, and accepted healthcare Digital Twin benchmarks are still limited. Patient-specific calibration is difficult, computational cost can be high, and uncertainty quantification is often incomplete.

Fourth, explainability, privacy, cybersecurity, consent, ownership, fairness, unequal access, responsibility, and regulation remain unresolved. These issues are not secondary concerns because Digital Twins use sensitive longitudinal data and may influence clinical interpretation. The proposed PFE does not claim to solve all these gaps; it explores a subset related to cardiovascular data acquisition, patient-state fusion, ECG diagnostic assistance, controlled what-if simulation, visualization, and human-supervised decision support.

6.6 Conclusion

This chapter compared selected Digital Twin studies and extracted conceptual, data-related, modeling, architectural, clinical, cardiovascular, and evaluation findings. The synthesis established the layered structure, cross-cutting requirements, data-flow logic, maturity distinctions, and validation principles required for trustworthy eHealth Digital Twins. The final chapter summarizes the contributions and limitations of the Master study and introduces CardioTwin as a practical cardiovascular specialization of these general findings.

Part IV

General Conclusion and PFE Transition

CHAPTER 7

GENERAL CONCLUSION AND PROPOSED METHOD FOR THE PFE

7.1 General Conclusion

This Master thesis studied Digital Twin technology in the context of eHealth. It showed that Digital Twins can support the transition from fragmented digital health applications toward dynamic, personalized, and service-oriented healthcare systems. A Digital Twin is not only a digital copy or a visualization tool; it is a connected virtual representation that uses data, models, analytics, simulation, and services to support monitoring, interpretation, prediction, and decision-making.

The study first clarified the conceptual foundations of Digital Twins, including the distinction between digital models, digital shadows, Digital Twins, and intelligent Digital Twins. It then introduced the eHealth and digital-health environment in which these concepts must be adapted. The healthcare-oriented review showed that Digital Twins for Health are receiving growing attention in personalized medicine, remote monitoring, hospital processes, public health, and cardiovascular applications. However, the field remains emerging, and many works are still conceptual, prototype-based, or limited to specific scenarios.

The comparative analysis produced a synthesis of the main findings from the structured literature review. It derived a general multi-layer eHealth Digital Twin architecture, identified cross-cutting requirements, described the data and feedback flow, and proposed evaluation criteria for trustworthy systems. The logic of the thesis is therefore: general Digital Twin foundations, eHealth context, healthcare applications, structured literature comparison, synthesized architecture and evaluation requirements, cardiovascular deep-dive, and finally the CardioTwin PFE specialization.

7.2 Main Contributions of the Master Study

The main contributions of this Master study are summarized as follows:

- clarification of Digital Twin concepts, maturity levels, and enabling technologies;

- analysis of eHealth foundations, including connected devices, health data, interoperability, and digital-health services;
- review of healthcare Digital Twin applications, with attention to cardiovascular examples;
- comparative study of selected general, healthcare-oriented, architecture-oriented, ethical, and cardiovascular works;
- synthesis of a general layered architecture for eHealth Digital Twins from recurring elements in the literature;
- identification of evaluation requirements covering technical, model, clinical, workflow, ethical, and governance dimensions;
- identification of research gaps related to synchronization, interoperability, validation, explainability, security, privacy, and clinical integration;
- preparation of a theoretical foundation for the CardioTwin PFE.

7.3 Limitations of the Study

This thesis is mainly a literature-based state-of-the-art and conceptual synthesis. It does not provide a full clinical deployment, a certified medical device, or a prospective clinical validation. The terminology of the field is heterogeneous, and published works do not always use the term Digital Twin with the same level of maturity. Some systems described in the literature are closer to digital models, simulations, or digital shadows than to fully synchronized and clinically integrated Digital Twins.

Another limitation is the limited number of mature real-world healthcare Digital Twin deployments. The field evolves rapidly, and the synthesis depends on the evidence available in the reviewed literature. Finally, the general architecture derived in Chapter 6 still requires use-case-specific implementation, data selection, model design, regulatory analysis, and validation before it can support clinical practice.

7.4 From the Master Study to the PFE

The Master thesis studied Digital Twins for eHealth from a general perspective. It analyzed healthcare applications and architectures, extracted general architectural and evaluation findings, and identified the requirements for trustworthy Digital Twin-based systems. The PFE builds on this foundation by operationalizing part of the synthesis in one clinical domain. It should therefore be understood as a practical specialization of the Master findings, not as proof that all unresolved research gaps have been solved.

The selected specialization is cardiovascular health. The cardiovascular domain satisfies the technical and methodological criteria identified in the literature synthesis: it requires longitudinal and personalized monitoring; it provides electrical, hemodynamic, clinical, physiological, and imaging information; patient presentations are heterogeneous; disease trajectories change over time; and treatment response varies between patients [1, 2]. The reviewed literature provides examples ranging from numerical simulation to patient-specific calibrated models and AI-assisted interpretation [23, 24]. This makes the domain a realistic pathway for an academic prototype while still requiring careful clinical validation before real medical deployment.

7.5 Introduction to the PFE Project

The proposed PFE is titled *CardioTwin – Design and Deployment of an AI-Powered Cardiovascular Digital Twin for Diagnostic Assistance and Therapeutic Simulation*. CardioTwin is introduced as a practical cardiovascular specialization of the general eHealth Digital Twin architecture synthesized in this Master study.

CardioTwin focuses on a subset of the general architecture derived in Chapter 6. It does not attempt to implement every layer and requirement at full clinical scale. Instead, it concentrates on a patient-oriented cardiovascular prototype in which ECG signals, systolic blood pressure, diastolic blood pressure, heart rate, demographic information, clinical history, symptoms, and other physiological measurements can provide complementary views of the patient's state. Clinicians may benefit from integrated diagnostic assistance, longitudinal follow-up, risk summaries, and controlled what-if simulations that explore possible therapeutic scenarios.

The objective of the PFE is to design and implement a patient-oriented cardiovascular Digital Twin platform that combines real biomedical measurements, ECG signal analysis, clinical and physiological data, AI-based diagnostic assistance, patient-state representation, cardiovascular risk assessment, therapeutic what-if simulation, monitoring, and visualization services.

7.6 Proposed PFE Methodology

The proposed PFE methodology follows a sequence derived from the general architecture and adapted to the cardiovascular use case.

7.6.1 Real-World Data Acquisition

The first step consists of acquiring cardiovascular and clinical data. The data may include ECG signals, systolic blood pressure, diastolic blood pressure, heart rate when available, patient demographic information, clinical history, symptoms, and other relevant physiological measurements. In the practical prototype, connected medical sensors and the MySignals eHealth development kit may be used to support biomedical data acquisition, depending on availability and implementation constraints.

7.6.2 Data Preprocessing and Validation

Acquired data must be prepared before they can update the twin or feed diagnostic modules. This step includes ECG filtering, artifact handling, normalization, segmentation, measurement validation, missing-data handling, and timestamp alignment. The purpose is to reduce noise, identify unreliable measurements, and preserve the temporal coherence required for longitudinal monitoring.

7.6.3 ECG Diagnostic-Assistance Module

The ECG module uses deep-learning models to analyze ECG signals for rhythm or abnormality detection. Biomedical datasets such as PTB-XL, MIT-BIH, SVDB, AFDB, or INCART may be used during development or evaluation, depending on the selected tasks and available resources. The goal is to provide diagnostic assistance, not autonomous diagnosis.

7.6.4 Clinical and Physiological Assessment

The system also integrates broader patient information, including age, sex, systolic blood pressure, diastolic blood pressure, clinical history, symptoms, and relevant cardiac measurements. These variables support a more complete cardiovascular assessment than ECG classification alone.

7.6.5 Patient-State Fusion

CardioTwin should fuse ECG findings, clinical risk indicators, physiological measurements, and historical information into a unified virtual representation of the patient's cardiovascular state. This makes the twin more than an isolated ECG classifier. The patient state should preserve the distinction between measured values, calculated values, model-estimated values, predicted values, and simulated values.

7.6.6 Digital Twin Update

The cardiovascular twin is updated when new validated patient information is received. Measured values may come directly from sensors or clinical records. Calculated values may be produced from formulas or preprocessing steps. Model-estimated values may be inferred from incomplete or indirect observations. Predicted values may represent expected future trends. Simulated values may represent hypothetical scenarios. These categories must not be confused because they have different levels of certainty and clinical meaning.

7.6.7 What-If Therapeutic Simulation

The what-if module is an exploratory clinical decision-support component. It may simulate changes in blood pressure, modifications of physiological parameters, lifestyle scenarios, treatment or medication-effect scenarios, and possible evolution over a selected time horizon. The generated scenarios are decision-support estimates based on models, evidence, or population-level effects and must not be interpreted as guaranteed individual clinical outcomes or autonomous medical prescriptions.

7.6.8 Service and Visualization Layer

The platform should provide patient dashboards, ECG visualization, historical trends, risk summaries, alerts, what-if scenario outputs, and clinician-oriented reports. These services must present information clearly and must distinguish observations, predictions, and simulations.

7.6.9 Human-in-the-Loop Decision Support

CardioTwin assists healthcare professionals but does not replace them. Clinical decisions remain under professional supervision. Simulated recommendations require medical interpretation, and autonomous therapeutic control is outside the scope of the PFE.

7.7 Evaluation Strategy

The PFE evaluation strategy should cover the main components of the prototype. Data acquisition evaluation will verify correct sensor communication, completeness, timestamps, signal quality, and measurement consistency. ECG-model evaluation will use metrics such as accuracy, precision, recall, F1-score, macro-F1, sensitivity, specificity, confusion matrices, and external dataset testing when available.

Patient-state evaluation will examine consistency between measurements and the stored state, correct longitudinal updates, and the handling of missing or contradictory measurements. What-if evaluation will focus on traceability of assumptions, consistency with cited medical evidence, plausible direction and magnitude of simulated effects, transparent uncertainty, warnings, and the absence of unsupported claims about individual treatment efficacy.

System evaluation will consider latency, modularity, reliability, usability, security, interoperability, and dashboard clarity. Full clinical evaluation would require clinician review, prospective assessment, ethical approval where necessary, representative patient data, and regulatory consideration. Such full validation remains outside the guaranteed scope of the academic prototype and should be treated as future work unless it is explicitly performed.

7.8 Future Work

Future work should first focus on implementing the CardioTwin prototype and integrating real biomedical sensors. Further work should improve the fusion of ECG findings with clinical and physiological data, validate patient-state updates, strengthen what-if simulations, add explainability and uncertainty communication, and integrate interoperability standards such as HL7 FHIR where appropriate.

Additional research should include clinician-centered usability evaluation, stronger cybersecurity and consent mechanisms, prospective validation, and a clearer path toward regulatory and ethical assessment. In a broader perspective, the same methodology could later be extended to other healthcare Digital Twin domains beyond cardiovascular monitoring.

7.9 Final Remarks

Digital Twins for eHealth represent a promising research direction at the intersection of healthcare, computer science, data engineering, artificial intelligence, simulation, and systems modeling. Their success will depend not only on technical performance, but also on trust, interoperability, clinical relevance, explainability, and responsible design. CardioTwin provides a practical continuation of this Master study by applying the general findings to a cardiovascular Digital Twin prototype.

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